

Practical No.9

Aim: Study & Implementation of Views.

Theory:

VIEW:

In SQL, a view is a virtual table based on the result-set of an SQL statement. A view contains rows and columns, just like a real table. The fields in a view are fields from one or more real tables in the database.

You can add SQL functions, WHERE, and JOIN statements to a view and present the data as if the data were coming from one single table.

A view is a virtual table, which consists of a set of columns from one or more tables. It is similar to a table but it does not store in the database. View is a query stored as an object.

It can fetch data from one or more tables and present it in a customized format, allowing developers to:

- Simplify Complex Queries: Encapsulate complex joins and conditions into a single object.
- Enhance Security: Restrict access to specific columns or rows.
- Present Data Flexibly: Provide tailored data views for different users.

Syntax:

```
CREATE VIEW <view_name> AS SELECT <set of fields>  
FROM relation_name WHERE (Condition);
```

Parameters:

- **view_name:** Name for the View
- **table_name:** Name of the table
- **condition:** Condition to select rows

Example:

```
SQL> CREATE VIEW employee AS SELECT empno,ename,job FROM EMP  
WHERE job = 'clerk';  
SQL> View created.
```

Example:

```
CREATE VIEW [Current Product List] AS  
SELECT ProductID, ProductName  
FROM Products  
WHERE Discontinued=No;
```

UPDATING A VIEW:

If you want to update the existing data within the view, use the **UPDATE** statement.

A view can be updated by using the following syntax:

```
UPDATE view_name
```

```
SET column1 = value1, column2 = value2..., columnN = valueN
```

```
WHERE [condition];
```

Note: Not all views can be updated using the UPDATE statement.

If you want to update the view definition without affecting the data, use the CREATE OR REPLACE VIEW statement. you can use this syntax

Syntax: CREATE OR REPLACE VIEW view_name AS

```
SELECT column_name(s)
```

```
FROM table_name
```

```
WHERE condition;
```

Rules to Update Views in SQL:

Certain conditions need to be satisfied to update a view. If any of these conditions are not met, the view can not be updated.

1. The SELECT statement which is used to create the view should not include GROUP BY clause or ORDER BY clause.
2. The SELECT statement should not have the DISTINCT keyword.
3. The View should have all NOT NULL values.
4. The view should not be created using nested queries or complex queries.
5. The view should be created from a single table. If the view is created using multiple tables then we will not be allowed to update the view.

DROPPING A VIEW:

A view can be deleted with the DROP VIEW command.

Syntax: DROP VIEW <view_name> ;

Uses of a View

A good database should contain views for the given reasons:

1. Restricting data access – Views provide an additional level of table security by restricting access to a predetermined set of rows and columns of a table.
2. Hiding data complexity – A view can hide the complexity that exists in multiple joined tables.
3. Simplify commands for the user – Views allow the user to select information from multiple tables without requiring the users to actually know how to perform a join.
4. Store complex queries – Views can be used to store complex queries.
5. Rename Columns – Views can also be used to rename the columns without affecting the base tables provided the number of columns in view must match the number of

columns specified in a select statement. Thus, renaming helps to hide the names of the columns of the base tables.

6. Multiple view facility – Different views can be created on the same table for different users.

Note: Write a query to be executed in lab with screen shots.

Result:

Thus, we implement the views in sql successfully.